INCH-POUND

MIL-T-29504/12
21 May 1992

MILITARY SPECIFICATION SHEET

TERMINI, FIBER OPTIC, CONNECTOR, REMOVABLE, ENVIRONMENT RESISTING, CLASS 5, TYPE II, STYLE A, PIN TERMINUS, FRONT RELEASE, STAINLESS STEEL (FOR MIL-C-28876 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-29504.

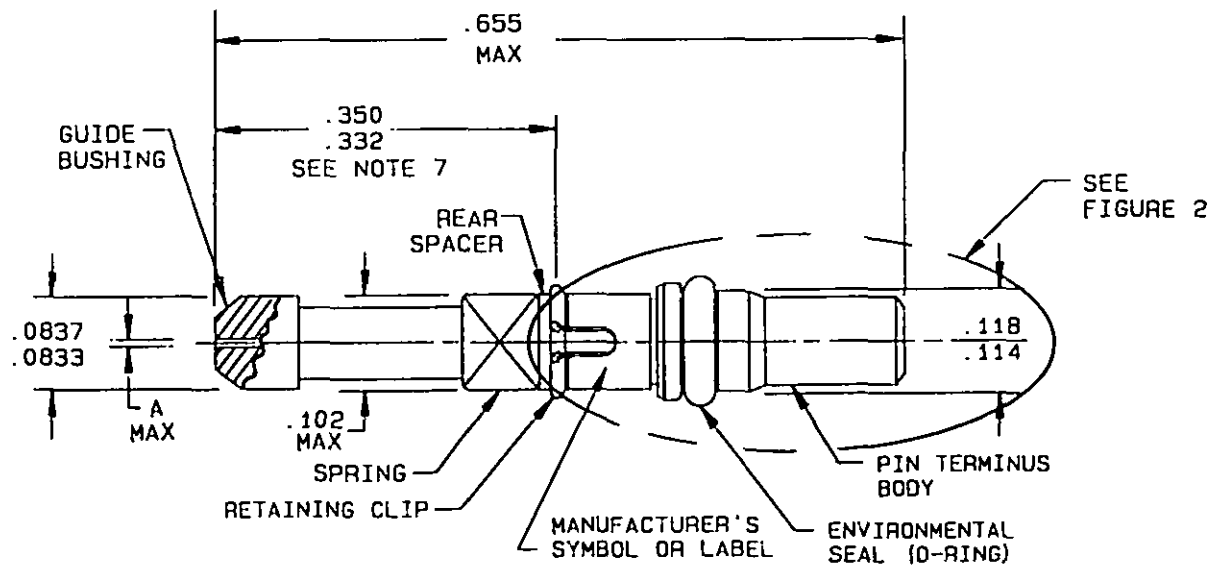


FIGURE 1. Pin terminus.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

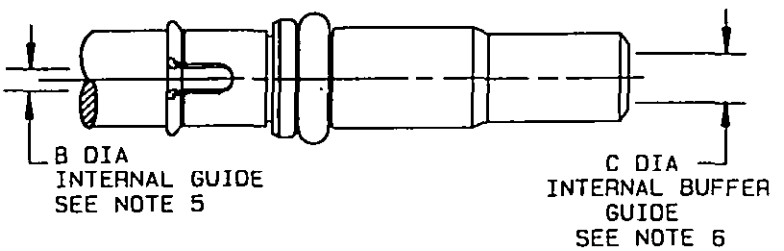
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Inches	mm
.001'	0.03
.0833	2.116
.0837	2.126
.101	2.57
.102	2.59
.104	2.64
.114	2.90
.118	3.00
.121	3.07
.125	3.17
.300	7.62
.332	8.43
.350	8.89
.655	16.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Guide bushing "A" diameter is based upon 1.00 inch = 25400.0 µm or 1.00 µm = 0.00003937 inch.
4. All diameters to be concentric within .002 (0.05 mm).
5. Dimensions apply after plating.
6. (O-ring) shall be a fluorosilicon material.
7. Dimension .332 (8.43 mm) or .350 (8.89 mm) to be measured when installed in connector insert or equivalent gauge fixture. See figure 3.
8. The TICC bands shall not appear on terminus. Individual bag or container shall have either TICC bands or equivalent MIL-T-29504 Part or Identifying Number (PIN) marking on container or a tag inside container.

FIGURE 1. Pin terminus - Continued.

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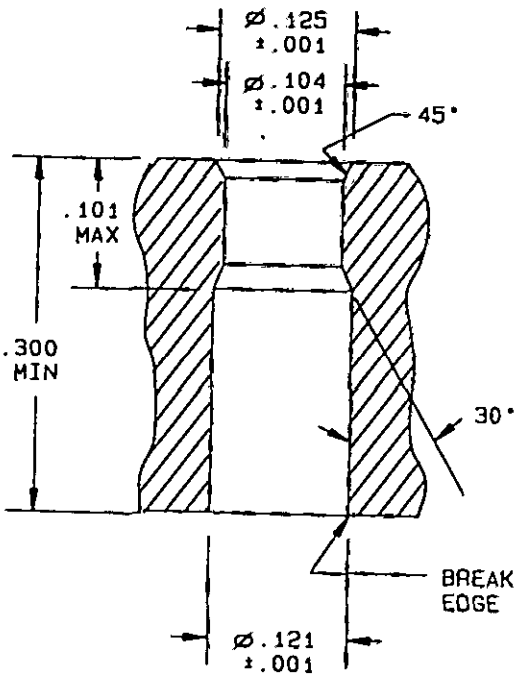


B diameter (internal guide)		C diameter (internal buffer guide)	
Inches	mm	Inches	mm
.017 .014	0.43 0.36	.044 .040	1.12 1.01

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within .002 (0.05 mm).
4. Dimensions shown after plating.
5. The "B" diameter of the internal guide is used to center coated optical waveguide fiber.
6. The "C" diameter of the internal buffer guide is used to center the optical waveguide fiber buffer.

FIGURE 2. Terminus internal buffer guide.

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Inches	mm
.001	0.03
.101	2.57
.104	2.64
.121	3.07
.125	3.18
.300	7.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within .002 (0.05 mm).
4. Dimensions shown after plating.

FIGURE 3. Insert equivalent fixture for terminus measurement.

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REQUIREMENTS:

Temperature: -55°C to +85°C operating, -62°C to +85°C storage.

Design and construction: See figures 1 and 2 and table I.

Weight: Not greater than 1.0 gram.

Adhesives: Use Hysol 1c, Epotek 330 or 353 series or equivalent as approved by the qualifying activity. Mixing and curing instructions shall be supplied with the assembly instructions provided with each terminus or applicable connector.

Material: Stainless steel (guide bushing), 300 series.

Tools: See table II.

Mating termini: MIL-T-29504/2 and /13.

Crimp sleeve: (2.5 mm) to be supplied with termini when specified in PIN.

The following MIL-T-29504 requirements are not applicable to this specification sheet:

Insertion loss	Optical discontinuities
Temperature life	Terminus retention
Terminus engagement and separation forces	Terminus cleaning
	Thermal shock
Shock	Vibration
Mating durability	Salt spray

PIN:

M29504/12	-XXXX	C
Basic specification and specification sheet	Dash number (TICC number, see table I)	Crimp sleeve (optional)

Qualification: To qualify or requalify termini to this specification sheet, all requirements of MIL-C-28876 shall be met using the qualifying terminus in a connector qualified to MIL-C-28876. Multimode termini shall be granted qualification upon the successful qualification of single mode termini.

Usage: Termini compliant with this specification sheet may be used in connectors other than MIL-C-28876 at the discretion of the acquiring activity.

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TABLE I. TICC numbers and dimensions for guide bushing.

TICC number		A diameter 1/	
Multimode	Single mode	Inches +0.0001 -0.0000	µm 2/
4087	4097	0.0048	122.0
4088	4098	0.0049	124.5
4089	4099	0.0050	127.0
4090	4100	0.0051	129.5
4091	4101	0.0054	137.0
4092	4102	0.0055	139.5
4093	4103	0.0056	142.0
4094	4104	0.0057	145.0
4095	4105	0.0091	231.5
4096	4106	0.0092	234.0

1/ For maximum performance, termini should be selected for the best (tightest) fit to the fiber. Fitting the next size larger terminus to a fiber may affect connector performance.
2/ Rounded to the nearest 0.5 µm.

TABLE II. Tools.

End size	Insertion tool	Removal tool	Polishing fixture tool
16	M81969/46-01 or M81969/49-01	M81969/47-01	M83523/6-01

Patent notice: The Government does not have a royalty-free license under the following patents for the benefit of manufacturers of the item, either for the Government or for use in equipment to be delivered to the Government.

Patent number	Patent expiration date
US 4707068	11/17/2004

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS, MC
Air Force - 17, 19, 80, 90, 99

User activities:
Navy - OS, YD
Air Force - 13, 14

Preparing activity:
Navy - SH

Agent:
DLA - ES

(Project 6060-0096-1)

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